



SIMATS Online Programming Lab

C PROGRAMMING



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CO5 - AT3 - HACKATHON / MINI PROJECT

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QUESTIONS

Q1

CO5 - AT3 - HACKATHON /
MINI PROJECT
100 marks



1/1 answered

Q1 CO5 - AT3 - HACKATHON / MINI PROJECT

100 marks

CO5 – AT3_HACKATHON QUESTIONS

1. A satellite continuously sends telemetry packets containing temperature, battery level, fuel status, altitude, and communication status to a ground station. Thousands of packets may arrive simultaneously from different processing channels. The system must store incoming records dynamically because the number of packets is unknown in advance.

Develop a C-based Real-Time Satellite Telemetry Processing System that:

- Uses dynamic memory allocation for telemetry records.
- Uses structures and pointers to manage satellite data.
- Uses functions to modularize packet processing.
- Uses POSIX threads to process multiple telemetry streams concurrently.
- Uses mutex/semaphore synchronization to prevent simultaneous updates to shared telemetry data.
- Stores processed telemetry in a file for later analysis.
- Uses command-line arguments to specify satellite ID and input/output files.
- Uses IPC or socket programming to simulate communication between the satellite simulator and ground station.

Hackathon Challenge: The system must continue processing telemetry even when multiple packets arrive at almost the same time without losing or corrupting records.

2. A hospital emergency department receives patient requests from multiple registration counters simultaneously. Each patient record contains patient ID, emergency level, symptoms, assigned doctor, and

treatment status.

Develop a Smart Emergency Patient Coordination System in C that:

- Dynamically allocates memory for patients.
- Uses functions for registration, search, update, and discharge.
- Uses pointers to modify patient records.
- Uses threads to simulate multiple hospital departments.
- Uses mutexes to prevent two departments from updating the same patient simultaneously.
- Stores patient history in files.
- Uses command-line arguments to configure the hospital department.
- Uses IPC to transfer emergency information between registration and treatment modules.

Hackathon Challenge: Guarantee that one patient's record cannot be overwritten when two departments update the same patient simultaneously.

3. A railway reservation system has multiple booking counters and online users attempting to reserve seats simultaneously. The number of seats is dynamically allocated according to the selected train. Two users may attempt to reserve the same seat at nearly the same time. Develop a Concurrent Railway Reservation System in C that:

- Dynamically allocates seats.
- Uses structures and pointers for passenger and train information.
- Uses functions for booking, cancellation, refund, and seat search.
- Uses pthreads to simulate concurrent users.
- Uses mutexes/semaphores to prevent double booking.
- Maintains transaction records in files.
- Uses command-line arguments for train number, seat count, and input file.
- Uses sockets to simulate communication between booking clients and a central reservation server.

Hackathon Challenge: If User A and User B request Seat 25 at exactly the same time, design synchronization so exactly one transaction succeeds and the other receives a failure/refund response.

4. A bank has multiple ATMs connected to a central transaction-processing server. Customers may withdraw, deposit, or transfer money simultaneously. A serious problem occurs when two ATMs access the same account balance simultaneously.

For example:

Initial Balance = ₹50,000

ATM 1 → Withdraw ₹40,000

ATM 2 → Withdraw ₹30,000

Develop a Concurrent Banking Transaction System in C using:

- Dynamic memory for account records.
- Pointers for balance updates.
- Functions for deposit, withdrawal, and transfer.
- pthreads for concurrent ATM simulation.
- Mutexes for account-level synchronization.
- Files for transaction logs.
- Command-line arguments for ATM configuration.
- Socket programming for ATM-to-server communication.

Hackathon Challenge: Design the system so that the bank account never reaches an invalid balance because of a race condition.

5. A large agricultural farm contains hundreds of sensor nodes measuring soil moisture, temperature, humidity, and water levels. The number of sensors may increase dynamically. Each sensor periodically sends information to a central monitoring system.

Develop a Smart Farm Monitoring System in C that:

- Dynamically creates sensor records.
- Uses pointers for sensor data manipulation.
- Uses functions for sensor registration and analysis.
- Uses recursive functions to traverse a hierarchical farm/field structure.
- Uses threads to simulate simultaneous sensor transmission.
- Uses synchronization to protect shared sensor data.
- Stores historical sensor readings in files.
- Uses sockets for sensor-to-server communication.
- Uses command-line arguments to specify field and sensor configuration.

Hackathon Challenge: Identify inactive sensors and automatically generate an alert when no data has been received within a specified interval.

6. A smart manufacturing plant contains hundreds of machines. Each machine produces vibration, temperature, pressure, and operating-time data. The monitoring system must detect abnormal machine behavior before failure occurs.

Develop a C-based Predictive Maintenance System that:

- Dynamically allocates memory for machines and sensor readings.
- Uses functions for data collection and analysis.
- Uses pointers to update machine states.
- Uses recursion to analyze hierarchical machine/subsystem information.

- Uses pthreads to process sensor streams concurrently.
- Uses mutexes to protect shared machine-status information.
- Stores historical data in files.
- Uses command-line arguments for threshold configuration.
- Uses IPC between sensor-processing and alert-generation modules.

Hackathon Challenge: Detect a machine whose temperature and vibration simultaneously exceed their thresholds and generate an emergency maintenance notification.

7. During a natural disaster, multiple field stations send emergency messages to a central control room. Messages may include location, severity, number of affected people, medical requirements, and rescue requirements. Communication may occur simultaneously from multiple stations.

Develop a Real-Time Disaster Communication System in C that:

- Dynamically stores incoming emergency messages.
- Uses pointers for message management.
- Uses functions for message classification and processing.
- Uses pthreads for simultaneous station communication.
- Uses mutexes to protect the central message queue.
- Uses sockets for network communication.
- Uses IPC between message-processing modules.
- Maintains an emergency log file.
- Uses command-line arguments to configure the server port and log file.

Hackathon Challenge: If 100 emergency messages arrive simultaneously, ensure no message loss, no duplicate processing, and correct priority ordering.

8. An airport handles thousands of bags belonging to passengers on multiple flights. Bags move through check-in counters, security scanning, loading, and aircraft transfer. Multiple scanning stations update baggage status concurrently.

Develop a Real-Time Airport Baggage Tracking System in C that:

- Dynamically creates baggage records.
- Uses pointers to update baggage status.
- Uses modular functions for check-in, scanning, loading, and delivery.
- Uses pthreads to simulate multiple baggage scanners.
- Uses synchronization to prevent inconsistent baggage status.
- Uses files to maintain baggage history.
- Uses sockets to communicate between airport terminals.
- Uses command-line arguments to configure terminal ID and server

port.

Hackathon Challenge: A bag must never simultaneously appear as LOADED and MISSING in the central system because of concurrent updates. Design synchronization to maintain consistent baggage state.

9. A surveillance network consists of multiple monitoring stations. Each station sends information about detected objects to a central command system. The number of detected objects is unpredictable, and multiple stations can report the same object simultaneously. Develop a C-based Border Surveillance Processing System that:

- Dynamically allocates memory for detected objects.
- Uses structures and pointers.
- Uses functions for object registration and classification.
- Uses recursion to traverse hierarchical surveillance zones.
- Uses pthreads to process multiple station feeds.
- Uses synchronization to protect the central object database.
- Uses files to store surveillance logs.
- Uses sockets for station-to-command-center communication.
- Uses IPC to pass alerts between detection and response processes.

Hackathon Challenge: If two stations detect the same object at nearly the same time, identify the duplicate and maintain one consistent object record.

10. A telecom company operates thousands of 5G towers. Each tower continuously reports signal strength, bandwidth, connected users, temperature, and fault status. Multiple monitoring processes simultaneously receive and update tower information.

Develop a 5G Network Fault Detection and Recovery System in C that:

- Dynamically allocates tower records.
- Uses pointers for efficient updates.
- Uses functions for fault detection and recovery.
- Uses pthreads to process multiple network streams.
- Uses mutexes/semaphores for synchronization.
- Uses files for historical fault logs.
- Uses command-line arguments for server configuration.
- Uses socket programming for tower-to-control-center communication.
- Uses IPC between fault detection and recovery processes.

Hackathon Challenge: The system must identify a tower failure, generate an alert, record the event, and initiate recovery without blocking the processing of other healthy towers.

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File 2 (optional)

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